Kernel Kaggle đang chạy lúc bàn giao (2026-10-29 ~17:00 UTC)

Các kernel này được tạo bởi khung chat cũ. Output của chúng **không** tự về branch mới. Kéo về theo cách sau:

```
KAGGLE_API_TOKEN=$(cat ~/.kaggle/tok_<acc>) kaggle kernels status <acc>/<slug>
KAGGLE_API_TOKEN=$(cat ~/.kaggle/tok_<acc>) kaggle kernels output <acc>/<slug> -p out/<slug>
cp out/<slug>/results/*.json analysis/results/
```

- Kernel đã COMPLETE vẫn kéo được output.
- Một số kết quả đã có sẵn trong `analysis/results/` (cột "đã kéo").
- Những kernel có Boreas trong cột "đã kéo" là kết quả đã dùng odometry đã sửa.

Run	Kernel	Job	Đã kéo
bd50b	`synchan5704/hmm-bd50b-1`	hmm_fix.py boreas cosplace; hmm_fix.py boreas dinov2	có
bd50b	`huyenco/hmm-bd50b-2`	hmm_fix.py boreas megaloc	có
bd50b	`yuan1103/hmm-bd50b-3`	hmm_fix.py boreas salad	có
bd50b	`synchan5704/hmm-bd50b-4`	hmm_fix.py boreas mixvpr	có
bd50b	`huyenco/hmm-bd50b-5`	hmm_fix.py boreas netvlad	có
bfix	`nhatlong1103/hmm-bfix-1`	hmm_fix.py boreas megaloc; nav_demo.py boreas mixvpr; cert_seq.py boreas netvlad	có
bfix	`yeshenhi/hmm-bfix-2`	hmm_fix.py boreas salad; nav_demo.py boreas cosplace; cert_seq.py boreas dinov2	có
bfix	`nguynduylm/hmm-bfix-3`	hmm_fix.py boreas mixvpr; nav_demo.py boreas netvlad	chưa
bfix	`loinguyentran/hmm-bfix-4`	hmm_fix.py boreas cosplace; nav_demo.py boreas dinov2	chưa
bfix	`nhatlong1103/hmm-bfix-5`	hmm_fix.py boreas netvlad; cert_seq.py boreas megaloc	chưa
bfix	`yeshenhi/hmm-bfix-6`	hmm_fix.py boreas dinov2; cert_seq.py boreas salad	chưa
bfix	`nguynduylm/hmm-bfix-7`	nav_demo.py boreas megaloc; cert_seq.py boreas mixvpr	có
bfix	`loinguyentran/hmm-bfix-8`	nav_demo.py boreas salad; cert_seq.py boreas cosplace	có
enull	`nguynduylm/hmm-enull-1`	nav_demo.py boreas megaloc	có
enull	`loinguyentran/hmm-enull-2`	nav_demo.py boreas salad	có
enull	`yuan1103/hmm-enull-3`	nav_demo.py 4s:countryside cosplace	có

Run	Kernel	Job	Đã kéo
fix6	`nhatlong1103/hmm-fix6-1`	hmm_fix.py cmu megaloc; hmm_fix.py 4s:old_town megaloc; hmm_fix.py 4s:countryside megaloc	có
fix6	`huyenco/hmm-fix6-2`	hmm_fix.py cmu salad; hmm_fix.py 4s:old_town salad; hmm_fix.py 4s:countryside salad	có
fix6	`loinguyentran/hmm-fix6-3`	hmm_fix.py cmu mixvpr; hmm_fix.py 4s:old_town mixvpr; hmm_fix.py 4s:countryside mixvpr	có
fix6	`mucnokieu/hmm-fix6-4`	hmm_fix.py cmu cosplace; hmm_fix.py 4s:old_town cosplace; hmm_fix.py 4s:countryside cosplace	có
fix6	`nguynduylm/hmm-fix6-5`	hmm_fix.py cmu netvlad; hmm_fix.py 4s:old_town netvlad; hmm_fix.py 4s:countryside netvlad	có
fix6	`synchan5704/hmm-fix6-6`	hmm_fix.py cmu dinov2; hmm_fix.py 4s:old_town dinov2; hmm_fix.py 4s:countryside dinov2	có
fix6	`yeshenhi/hmm-fix6-7`	hmm_fix.py nord megaloc; hmm_fix.py 4s:office_loop megaloc; hmm_fix.py rc megaloc	có
fix6	`yuan1103/hmm-fix6-8`	hmm_fix.py nord salad; hmm_fix.py 4s:office_loop salad; hmm_fix.py rc salad	có
fix6	`nhatlong1103/hmm-fix6-9`	hmm_fix.py nord mixvpr; hmm_fix.py 4s:office_loop mixvpr; hmm_fix.py rc mixvpr	có
fix6	`huyenco/hmm-fix6-10`	hmm_fix.py nord cosplace; hmm_fix.py 4s:office_loop cosplace; hmm_fix.py rc cosplace	có
fix6	`loinguyentran/hmm-fix6-11`	hmm_fix.py nord netvlad; hmm_fix.py 4s:office_loop netvlad; hmm_fix.py rc netvlad	có
fix6	`mucnokieu/hmm-fix6-12`	hmm_fix.py nord dinov2; hmm_fix.py 4s:office_loop dinov2; hmm_fix.py rc dinov2	có
fix7	`nhatlong1103/hmm-fix7-1`	hmm_fix.py boreas megaloc; hmm_fix.py nord mixvpr; hmm_fix.py 4s:office_loop netvlad	chưa
fix7	`huyenco/hmm-fix7-2`	hmm_fix.py boreas salad; hmm_fix.py nord cosplace; hmm_fix.py 4s:office_loop dinov2	chưa

Run	Kernel	Job	Đã kéo
fix7	`loinguyentran/hmm-fix7-3`	hmm_fix.py boreas mixvpr; hmm_fix.py nord netvlad; hmm_fix.py 4s:old_town megaloc	chưa
fix7	`mucnokieu/hmm-fix7-4`	hmm_fix.py boreas cosplace; hmm_fix.py nord dinov2; hmm_fix.py 4s:old_town salad	chưa
fix7	`nguynduylm/hmm-fix7-5`	hmm_fix.py boreas netvlad; hmm_fix.py rc megaloc; hmm_fix.py 4s:old_town mixvpr	chưa
fix7	`synchan5704/hmm-fix7-6`	hmm_fix.py boreas dinov2; hmm_fix.py rc salad; hmm_fix.py 4s:old_town cosplace	chưa
fix7	`yeshenhi/hmm-fix7-7`	hmm_fix.py cmu megaloc; hmm_fix.py rc mixvpr; hmm_fix.py 4s:old_town netvlad	chưa
fix7	`yuan1103/hmm-fix7-8`	hmm_fix.py cmu salad; hmm_fix.py rc cosplace; hmm_fix.py 4s:old_town dinov2	chưa
fix7	`nhatlong1103/hmm-fix7-9`	hmm_fix.py cmu mixvpr; hmm_fix.py rc netvlad; hmm_fix.py 4s:countryside megaloc	chưa
fix7	`huyenco/hmm-fix7-10`	hmm_fix.py cmu cosplace; hmm_fix.py rc dinov2; hmm_fix.py 4s:countryside salad	chưa
fix7	`loinguyentran/hmm-fix7-11`	hmm_fix.py cmu netvlad; hmm_fix.py 4s:office_loop megaloc; hmm_fix.py 4s:countryside mixvpr	chưa
fix7	`mucnokieu/hmm-fix7-12`	hmm_fix.py cmu dinov2; hmm_fix.py 4s:office_loop salad; hmm_fix.py 4s:countryside cosplace	chưa
fix7	`nguynduylm/hmm-fix7-13`	hmm_fix.py nord megaloc; hmm_fix.py 4s:office_loop mixvpr; hmm_fix.py 4s:countryside netvlad	chưa
fix7	`synchan5704/hmm-fix7-14`	hmm_fix.py nord salad; hmm_fix.py 4s:office_loop cosplace; hmm_fix.py 4s:countryside dinov2	chưa
navF	`yuan1103/hmm-navF-1`	nav_demo.py boreas megaloc; nav_demo.py boreas dinov2; nav_demo.py cmu netvlad	chưa
navF	`yeshenhi/hmm-navF-2`	nav_demo.py boreas salad; nav_demo.py cmu megaloc; nav_demo.py cmu dinov2	chưa

Run	Kernel	Job	Đã kéo
navF	`nhatlong1103/hmm-navF-3`	nav_demo.py boreas mixvpr; nav_demo.py cmu salad	chưa
navF	`nhatlong1103/hmm-navF-4`	nav_demo.py boreas cosplace; nav_demo.py cmu mixvpr	push lỗi, đã chạy lại
navF	`yuan1103/hmm-navF-5`	nav_demo.py boreas netvlad; nav_demo.py cmu cosplace	chưa
navFc	`yuan1103/hmm-navFc-1`	nav_demo.py nord megaloc; nav_demo.py 4s:office_loop mixvpr; nav_demo.py 4s:old_town netvlad	có
navFc	`yeshenhi/hmm-navFc-2`	nav_demo.py nord salad; nav_demo.py 4s:office_loop cosplace; nav_demo.py 4s:old_town dinov2	có
navFc	`huyenco/hmm-navFc-3`	nav_demo.py nord mixvpr; nav_demo.py 4s:office_loop netvlad; nav_demo.py 4s:countryside megaloc	chưa
navFc	`nhatlong1103/hmm-navFc-4`	nav_demo.py nord cosplace; nav_demo.py 4s:office_loop dinov2; nav_demo.py 4s:countryside salad	chưa
navFc	`yuan1103/hmm-navFc-5`	nav_demo.py nord netvlad; nav_demo.py 4s:old_town megaloc; nav_demo.py 4s:countryside mixvpr	chưa
navFc	`yeshenhi/hmm-navFc-6`	nav_demo.py nord dinov2; nav_demo.py 4s:old_town salad; nav_demo.py 4s:countryside cosplace	chưa
navFc	`huyenco/hmm-navFc-7`	nav_demo.py 4s:office_loop megaloc; nav_demo.py 4s:old_town mixvpr; nav_demo.py 4s:countryside netvlad	chưa
navFc	`nhatlong1103/hmm-navFc-8`	nav_demo.py 4s:office_loop salad; nav_demo.py 4s:old_town cosplace; nav_demo.py 4s:countryside dinov2	chưa
navG	`synchan5704/hmm-navG-1`	nav_demo.py boreas cosplace	chưa
navG	`loinguyentran/hmm-navG-2`	nav_demo.py cmu mixvpr	chưa

Thứ tự quan trọng: `pull_kernels.py` chép theo thứ tự bảng. Run sau ghi đè run trước:

- fix7 ghi đè fix6 (null cũ);
- navF, navFc và navG ghi đè enull và bfix (1 seed hoặc không có biến thể p_out).

Kernel có trạng thái "push lỗi" đã được chạy lại ở run khác: navF-4 thành navG.